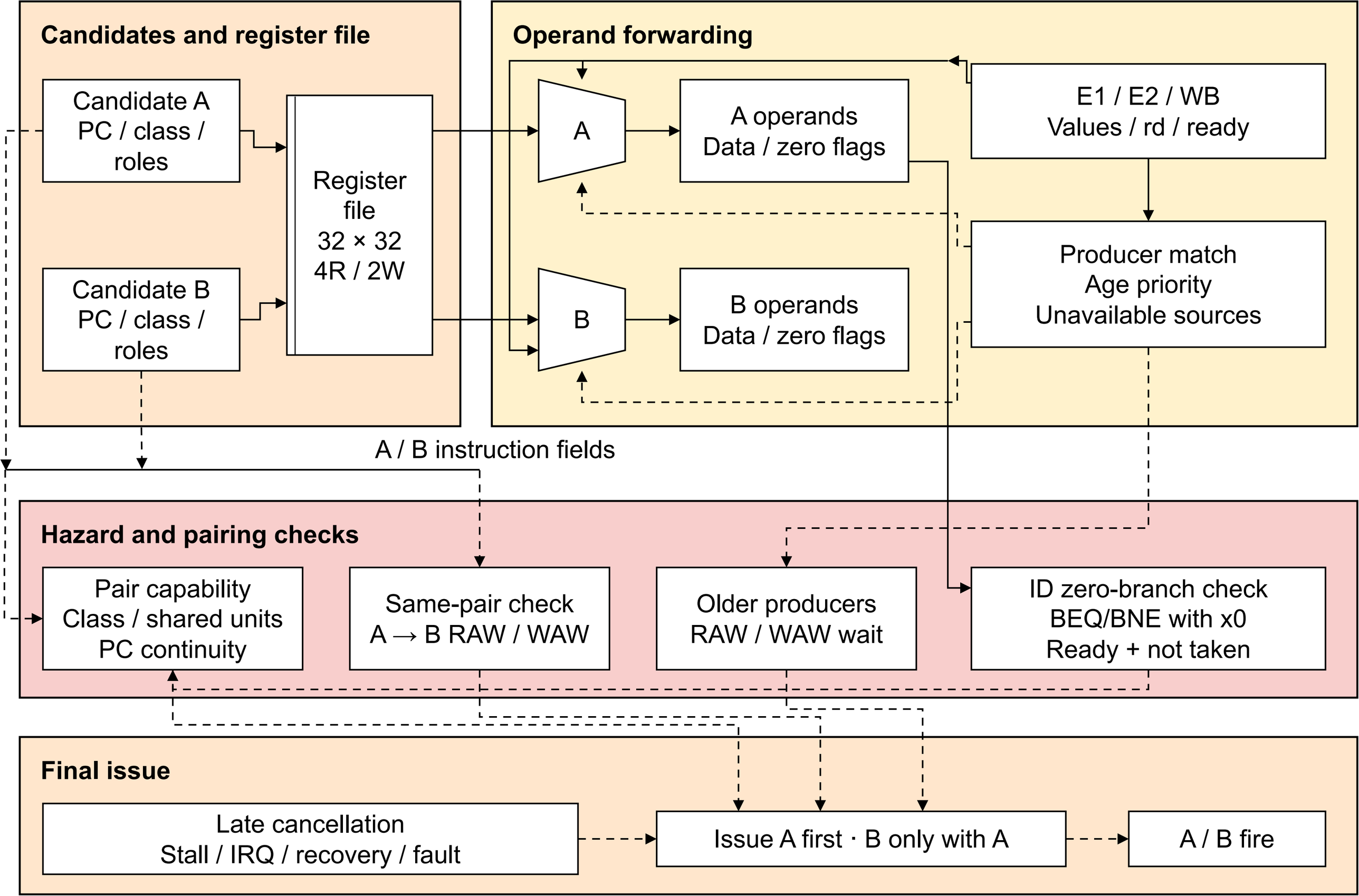
Candidates and register file
Candidate A
PC / class /
roles
Candidate B
PC / class /
roles
Register
file
32 × 32
4R / 2W
Operand forwarding
A
B
A operands
Data / zero flags
B operands
Data / zero flags
E1 / E2 / WB
Values / rd / ready
Producer match
Age priority
Unavailable sources
A / B instruction fields
Hazard and pairing checks
Pair capability
Class / shared units
PC continuity
Same-pair check
A → B RAW / WAW
Older producers
RAW / WAW wait
ID zero-branch check
BEQ/BNE with x0
Ready + not taken
Final issue
Late cancellation
Stall / IRQ / recovery / fault
Issue A first · B only with A
A / B fire